Implementation Plan Report

Kuwait National Environmental Data Management Plan (NEDMP)

Implementation Plan

A practical, phased and measurable mechanism for operationalizing environmental data governance, quality, interoperability, access and sustainability.

Document status	Implementation-ready draft for national review
Implementation horizon	Five-year phased implementation cycle
Evidence base	Eight NEDMP source reports and extracted source-evidence matrix
Prepared style	UNEP-style professional one-column report with updateable Word fields

Prepared for senior national decision-makers, EPA/KEPA coordination functions and participating institutions.

Publication Information

This document translates the strategic direction of the National Environmental Data Management Plan into an operational implementation architecture for the State of Kuwait. It is based primarily on the attached NEDMP source reports and does not assign legal duties, budgets or institutional mandates unless supported by the source evidence and subject to national approval.

Institutional names follow the source documents, which use Environment Public Authority, Kuwait Environment Public Authority, EPA and KEPA. The implementation plan uses Environment Public Authority (EPA/KEPA) where clarity is needed.

Disclaimer

Proposed committees, roles, financing arrangements, workflows and targets are implementation recommendations. They require validation by the competent national authorities, legal review where relevant and budget approval before they can be treated as formal government obligations.

Acknowledgements

The plan is built from national consultation, diagnostic assessment, benchmarking and governance framework material prepared under the Kuwait NEDMP process. It reflects the technical evidence provided through the eight source reports.

Executive Summary

The NEDMP Implementation Plan defines how Kuwait can move from a strategic environmental data governance framework to coordinated implementation. It distinguishes the strategic NEDMP from this implementation plan and from the future detailed annual action plans. The strategic document defines the vision and objectives; this plan defines implementation arrangements, workstreams, sequencing, indicators, risks and sustainability mechanisms; the action plan should later translate these into named annual activities and resource allocations.

The evidence base indicates that Kuwait has a substantial environmental data production base. Overall data availability is reported at 70.9 per cent, with a 29.1 per cent gap, interoperability readiness at 74.1 per cent and data quality readiness at 73.4 per cent. Governance readiness, however, is lower at 57.6 per cent. The implementation logic therefore starts with governance, roles, standards and classification before moving to platform integration and advanced analytics.

Implementation is organized around five workstreams aligned with the approved NEDMP foundational pillars: Data Governance and Institutional Management; Data Quality; Data Interoperability and System Integration; Data Sharing, Accessibility and Transparency; and Capacity Development and Sustainability. These workstreams operate as an integrated programme, not separate projects.

The plan proposes a national oversight and coordination structure, a Steering Committee, a National Environmental Data Governance Committee, an EPA/KEPA secretariat and implementation unit, thematic technical working groups, institutional focal points and formal data owner, steward and custodian roles. These are labelled as proposed arrangements subject to national approval.

The revised one-column edition also adds a national-alignment section showing how the implementation plan supports Kuwait Vision 2035, the Kuwait National Development Plan and Kuwait Environmental Protection Law No. 42 through governed datasets, classification, quality controls, interoperability, access rules and measurable review.

The implementation horizon is five years. The sequence moves from institutional endorsement, rules and standards, dataset inventory and classification, priority integration pilots, portal services and open-data pathways, to institutionalization, adaptive review and preparation of the next cycle.

Indicator	Value	Implementation interpretation	Data bar
Overall data availability	70.9%	Build on an active production base while closing documented gaps.	||||||| 70.9%
Environmental data gap	29.1%	Prioritize dataset inventory, gap closure and recurring updates.	||| 29.1%
Interoperability readiness	74.1%	Move from active exchange to governed, machine-readable recurrent flows.	||||||| 74.1%
Data quality readiness	73.4%	Convert partial quality practice into mandatory QA/QC and metadata controls.	||||||| 73.4%

Governance readiness	57.6%	Treat governance, stewardship, decision rights and SOPs as first implementation priority.	|||||| 57.6%

Figure 1. Source-derived readiness dashboard for NEDMP implementation

Table of Contents

List of Acronyms, Abbreviations and Report Codes

Acronym	Full name
A/R	RACI role code meaning Accountable and Responsible
A/R/C/I	RACI table role codes: Accountable / Responsible / Consulted / Informed
AI	Artificial intelligence
API	Application programming interface
CSB	Central Statistical Bureau
DAMA	Data Management Association International
DAMA-DMBOK	Data Management Body of Knowledge
DMBOK	Data Management Body of Knowledge
DR	Disaster recovery
EC	European Commission
eMISK	Environmental Monitoring Information System of Kuwait
EPA	Environment Public Authority
EPA/KEPA	Environment Public Authority / Kuwait Environment Public Authority
FAIR	Findable, Accessible, Interoperable and Reusable
FDES	Framework for the Development of Environment Statistics
GEDS	Global Environmental Data Strategy
GIS	Geographic information system
IEA	Integrated environmental assessment
INSPIRE	Infrastructure for Spatial Information in the European Community
ISO	International Organization for Standardization
ISO/IEC	International Organization for Standardization / International Electrotechnical Commission
IT	Information technology
KEPA	Kuwait Environment Public Authority, used in source documents alongside EPA
KEPA/UNEP	Kuwait Environment Public Authority / United Nations Environment Programme
KISR	Kuwait Institute for Scientific Research
KNDP	Kuwait National Development Plan
KPI	Key performance indicator
M&E	Monitoring and evaluation
MEA	Multilateral environmental agreement
MoU	Memorandum of understanding
NEDMP	National Environmental Data Management Plan
P1.1-P5.4	Implementation programme codes; P means Programme, the first number maps to the related workstream/pillar and the decimal number is the programme sequence
PMO	Programme management office
QA/QC	Quality assurance and quality control

RACI	Responsible, Accountable, Consulted and Informed
RBM	Results-based management
S1	Source citation code for Section I - National Vision, Objectives and Pillars of Environmental Data Governance
S2	Source citation code for Section II - Alignment with Key National and International Frameworks
S3	Source citation code for Section III - Benchmarking Best Practices in Environmental Data Governance
S4I	Source citation code for Section IV Part I - Current State of National Environmental Data
S4II	Source citation code for Section IV Part II - EPA Department Assessment
S4III	Source citation code for Section IV Part III - All-Stakeholder Assessment
S6	Source citation code for NEDMP Data Classification Governance Framework
SDG	Sustainable Development Goal
SDMX	Statistical Data and Metadata eXchange
SEEA	System of Environmental-Economic Accounting
SEIS	Shared Environmental Information System
SO	Strategic objective
SO1-SO5	NEDMP strategic objective codes 1 to 5
SOE	State of the Environment
SOP	Standard operating procedure
TOR	Terms of reference
UN	United Nations
UNEP	United Nations Environment Programme
WESR	World Environment Situation Room
WS	Workstream
WS1-WS5	NEDMP implementation workstream codes 1 to 5

List of Figures

List of Tables

List of Boxes

Chapter 1. Introduction

This chapter defines the purpose, scope, users and evidence base of the NEDMP Implementation Plan. It translates the strategic NEDMP into a coordinated implementation mechanism, while leaving detailed annual activities for the subsequent Action Plan.

1.1 Background

Kuwait's environmental data ecosystem is already active, but it remains distributed across institutions, systems, reporting duties and thematic domains. The NEDMP was prepared through consultation, diagnostic assessment and benchmarking to create a unified direction for environmental data governance, quality, interoperability, sharing and capacity development (Section I - National Vision, Objectives and Pillars of Environmental Data Governance, p. 6; NEDMP strategic document, p. 4).

The implementation challenge is not only technical. The source assessments show that data are produced and exchanged, but governance readiness is weaker than technical readiness. The plan therefore treats environmental data as a strategic national asset that requires roles, standards, quality controls, access rules and sustainability arrangements before full platform integration is scaled.

1.2 Purpose of the Implementation Plan

The purpose is to define how the NEDMP will be governed, coordinated, sequenced, operationalized, financed, monitored, evaluated, adapted, institutionalized and sustained. It is a bridge between the strategic NEDMP and the future detailed Action Plan.

1.3 Scope and intended users

The plan is intended for senior decision-makers, EPA/KEPA coordination functions, participating ministries and agencies, data owners, technical teams, statistical and planning entities, research institutions and partners involved in environmental data production, sharing, assessment and public information.

1.4 Relationship with the strategic NEDMP and Action Plan

The strategic NEDMP sets the vision, objectives, pillars and expected outcomes. This Implementation Plan defines institutional mechanisms, workstreams, programmes, implementation sequence, indicators, risks and sustainability arrangements. The Action Plan should later provide annual activities, named responsible entities, deadlines, approved resource allocations and procurement or technical work packages.

1.5 Implementation period

The recommended implementation horizon is five years, structured into five phases: mobilization, foundation, priority implementation, scale-up and adaptive renewal. The phasing is indicative and should be validated against national approval processes and budget cycles.

1.6 Methodology and evidence base

The methodology combined extraction from the eight source PDFs, synthesis of the NEDMP strategic framework, review of diagnostic findings, mapping of governance and classification recommendations, and conversion of those findings into implementation workstreams, programmes, matrices and indicators.

Box 1. Evidence note on diagnostic statistics

> The all-stakeholder assessment and the data classification governance framework report overall availability of 70.9 per cent, data gap of 29.1 per cent, interoperability readiness of 74.1 per cent, data quality readiness of 73.4 per cent and governance readiness of 57.6 per cent. These values are used consistently because they are the most complete implementation-oriented evidence set in the attached sources.
>
> Where earlier strategic text reports slightly different dashboard values, those are treated as earlier strategic mapping values rather than the implementation baseline.

Chapter 2. Strategic and Results Framework

This chapter preserves the approved strategic architecture and converts it into a results-based implementation framework.

2.1 Mandate and vision

The NEDMP mandate is grounded in Kuwait's environmental law, national development priorities and international environmental reporting needs. The vision is to create a trusted, integrated and accessible environmental data ecosystem that supports evidence-based environmental governance, public information, digital transformation and international reporting (Section I - National Vision, Objectives and Pillars of Environmental Data Governance, p. 7; Section II - Alignment with Key National and International Frameworks, p. 15; NEDMP strategic document, p. 6).

2.2 Strategic objectives and foundational pillars

Table 1. Approved strategic objectives retained in the Implementation Plan

Objective	Approved wording	Implementation meaning
SO1	Institutionalize Environmental Data Governance	Mandate, roles, decision rights, data ownership and accountability are formalized.
SO2	Ensure High-Quality and Standardized Environmental Data	Quality assurance, metadata, validation, provenance and common data definitions are applied.
SO3	Enable Interoperability and System Integration	Systems, platforms, formats and exchange mechanisms are aligned for recurrent data flow.
SO4	Strengthen Data Sharing, Accessibility and Transparency	Public, internal, restricted and confidential access pathways are governed and transparent.
SO5	Build Sustainable National Capacities	Human capacity, institutional capability, financing and continuous improvement are secured.

Table 2. NEDMP foundational pillars as implementation workstreams

Pillar	Approved wording	Operational role
Pillar 1	Data Governance and Institutional Management	Approves rules, assigns roles and operates national coordination.
Pillar 2	Data Quality	Makes metadata, QA/QC and traceability mandatory for priority datasets.
Pillar 3	Data Interoperability and System Integration	Creates common formats, APIs and system readiness pathways.
Pillar 4	Data Sharing, Accessibility and Transparency	Balances open data, controlled access, legal review and publication readiness.
Pillar 5	Capacity Development and Sustainability	Funds skills, staffing, change management and long-term operation.

2.3 Theory of Change

If Kuwait formalizes environmental data governance roles, classifies datasets, applies metadata and QA/QC, builds interoperable exchange pathways and invests in capacity and sustainability, then environmental data can become trusted, reusable and accessible for policy, assessment, enforcement, public information and international reporting.

Inputs	Activities	Outputs	Outcomes	Impact

Mandate, evidence, institutions, systems	Governance activation, standards, inventory, integration pilots	Rules, registry, classified datasets, portal modules, trained focal points	Reliable, interoperable, governed and accessible environmental data	Better environmental decisions, reporting, transparency and resilience

***Figure* 2. NEDMP theory of change**

2.4 Results hierarchy

***Table* 3. Results hierarchy for the NEDMP Implementation Plan**

Level	Definition	Illustrative content
Impact	Long-term national change	Environmental governance and policy decisions are supported by trusted national environmental data.
Strategic outcomes	Five objective-level results	Governance, quality, interoperability, access/transparency and capacity are improved.
Intermediate outcomes	Workstream-level changes	Data owners assigned, metadata complete, exchange pilots operating and open-data pathway active.
Outputs	Deliverables under programmes	Terms of reference (TORs), registries, standards, templates, portal modules, training records and review reports.
Activities	Detailed actions	To be specified in annual Action Plans.
Indicators	Performance measures	KPI framework, monitoring sheets and annual review.

2.5 Alignment with Kuwait Vision 2035, KNDP and Law 42

The implementation plan operationalizes the NEDMP mandate by connecting environmental data governance to Kuwait Vision 2035, the Kuwait National Development Plan (KNDP) and Kuwait Environmental Protection Law No. 42. The alignment is practical: national vision and development priorities define the policy value of environmental data, while Law 42 frames the legal need for monitoring, reporting, coordination, environmental information and safeguarded disclosure. The implementation mechanisms are the bridge: governance bodies, dataset registries, classification, QA/QC, exchange protocols, portal services, KPIs and annual review (Section II - Alignment with Key National and International Frameworks, p. 6; Section II - Alignment with Key National and International Frameworks, p. 15; NEDMP strategic document, p. 6; NEDMP Data Classification Governance Framework, p. 9).

This means the plan should not be read as a stand-alone technology programme. It is a national implementation instrument that helps Kuwait produce reliable evidence for sustainable development decisions, track environmental performance, strengthen institutional coordination and release information safely where law, classification and data quality allow.

National drivers	Implementation mechanisms	Operational outputs	Public value
Kuwait Vision 2035, KNDP and Law 42	Governance, classification, metadata, QA/QC, interoperability and monitoring	Registry, standards, portal services, KPI dashboard and annual review	Trusted evidence for planning, compliance, reporting, transparency and resilience

***Figure* 3. National alignment logic for the NEDMP implementation plan**

***Table* 4. Alignment of the NEDMP implementation plan with national policy and legal instruments**

Instrument	Relevant national direction	How the Implementation Plan responds	Evidence/output
Kuwait Vision 2035	Strengthens effective government, digital transformation, sustainable living environment and evidence-based planning.	Creates a governed national environmental data asset with clear roles, quality controls, interoperability and performance review.	Governance structure, dataset registry, KPI framework and national environmental data services.

Kuwait National Development Plan	Requires coordinated implementation, monitoring of development priorities and cross-institutional evidence for planning.	Links workstreams, annual action planning, responsible roles, risks, resources and KPIs into a five-year delivery architecture.	Implementation matrix, critical path, resource framework, monitoring and evaluation (M&E) cycle and annual review.
Kuwait Environmental Protection Law No. 42	Supports environmental protection, monitoring, reporting, coordination, data exchange and controlled public information.	Embeds legal/security review, classification, access workflow, metadata, QA/QC and documented release decisions before data sharing is scaled.	Classification model, access protocol, data-sharing templates, audit logs and publication-readiness checks.

Box 2. Implementation alignment rule

Every annual NEDMP Action Plan should state which Vision 2035, KNDP and Law 42 alignment point each activity supports, which dataset or institutional process it improves, and which KPI will show progress.

Activities that cannot be linked to a national priority, legal requirement, dataset improvement or measurable output should be reconsidered or deferred.

Chapter 3. Implementation Principles and Operating Model

The operating model is designed to keep governance ahead of technology, while ensuring that technology, data quality, transparency and sustainability reinforce each other.

3.1 Implementation principles

- National ownership and whole-of-government coordination.
- Environmental data as a strategic national asset.
- Governance before technology, with interoperability and quality by design.
- Security, privacy, data sovereignty and continuity by design.
- Open by default where legally and institutionally appropriate; controlled access where sensitivity requires it.
- Lifecycle management, reuse before duplication and once-only collection where feasible.
- Evidence-based decision-making, stakeholder participation and continuous improvement.

3.2 Operating model

The operating model has four layers: national governance, institutional delivery, technical data services and monitoring/adaptive management. Each layer must have clear decision rights and feedback loops. The Environmental Monitoring Information System of Kuwait (eMISK) is treated as an operational system to be integrated through governed metadata, quality and access rules.

National governance	Institutional delivery	Technical data services	Monitoring and learning
Oversight, steering, governance committee, approvals	Focal points, owners, stewards, custodians, working groups	Registry, metadata, QA/QC, classification, portal, APIs, eMISK links	KPIs, reviews, risk register, corrective actions

Figure 4. NEDMP implementation architecture

3.3 Governance-first implementation sequence

The sequence below shows how the implementation should move from institutional authority to technical integration and long-term institutionalization. The numbered stages are cumulative: each stage creates the conditions for the next one.

1	**Governance**	Approve roles, decision rights, focal points and escalation routes.
2	**Standards and quality**	Issue metadata, QA/QC, data dictionary and standard operating procedure requirements.
3	**Inventory and classification**	Register priority datasets, assign owners/stewards and apply access classification.
4	**Interoperability**	Define formats, controlled vocabularies, APIs and eMISK-linked exchange protocols.
5	**Access and public services**	Operate controlled access, publication-readiness checks and open-data releases.
6	**Analytics and AI readiness**	Use governed datasets for dashboards, gap analysis, responsible AI and reporting.
7	**Institutionalization**	Embed annual review, funding, training and continuous improvement into the next cycle.

Figure 5. Governance-first sequence for implementation

Chapter 4. Institutional Governance and Coordination Arrangements

This chapter proposes a coordination structure that distinguishes legal mandates, existing responsibilities and implementation roles requiring formal approval.

4.1 Proposed institutional structure

The source evidence indicates that institutional fragmentation, role ambiguity and weak coordination are major implementation risks. Proposed structures are therefore labelled as subject to national approval and should be formalized through appropriate government decisions, terms of reference and budget arrangements (Section IV Part II - EPA Department Assessment, p. 63; NEDMP Data Classification Governance Framework, p. 12).

Oversight	Strategic direction	Technical governance	Delivery network
High-level national oversight	NEDMP Steering Committee	National Environmental Data Governance Committee and technical working groups	EPA/KEPA secretariat, PMO, focal points, data owners, stewards and custodians

Figure 6. Proposed institutional governance structure

Table 5. Institutional roles and approval status

Body or role	Status	Core function
High-level national oversight mechanism	Proposed, subject to approval	Endorse priorities, resolve inter-ministerial issues and connect NEDMP to national planning.
NEDMP Steering Committee	Proposed, subject to approval	Provide strategic direction, approve annual workplans and review implementation performance.
National Environmental Data Governance Committee	Proposed, subject to approval	Approve standards, classification rules, dataset prioritization and data-sharing protocols.
EPA/KEPA secretariat and coordination function	Existing mandate plus proposed implementation function	Coordinate implementation, maintain the registry, prepare reporting and support committees.
Programme management office or technical implementation unit	Proposed, subject to approval	Manage delivery schedule, dependencies, risks, evidence matrix and progress reporting.
Thematic technical working groups	Proposed	Develop domain-specific metadata, QA/QC, indicators and integration requirements.
Institutional environmental data focal points	Proposed network	Coordinate within institutions, maintain dataset records and facilitate exchange.
Data owners	Existing responsibilities to be formalized	Approve classification, correction, sharing and publication decisions for assigned datasets.
Data stewards	Proposed operational role	Maintain metadata, quality checks, versioning, dataset status and issue logs.
Data custodians	Existing/proposed technical role	Operate repositories, backups, access logs and technical safeguards.
Legal and compliance functions	Existing functions connected to NEDMP	Review restricted/confidential releases, data-sharing agreements and legal constraints.
Cybersecurity and business continuity functions	Existing/proposed coordination	Apply security controls, backup, disaster recovery and continuity requirements.
National portal operating function	Proposed	Operate dataset registry, access workflows, publication services and integration modules.

4.2 Decision rights and escalation

Issue detected	Institutional review	Technical working group	Governance committee	Steering escalation
Quality gap, access request, conflict, delay or legal concern	Data owner/steward/focal point logs and assesses	Technical group proposes resolution	Governance committee approves cross-institutional decision	Steering Committee resolves policy or resource issue

Figure 7. Decision-making and escalation flow

Legend: RACI means Responsible, Accountable, Consulted and Informed. In the table, A means accountable, R means responsible, C means consulted, I means informed, and A/R means the same actor is both accountable and responsible.

Table 6. RACI model for core implementation decisions

Activity	Steering	EPA/KEPA secretariat	Data owner/steward	Legal/security	Public/users
Approve NEDMP implementation governance	A	R	C	C	I
Maintain source-evidence matrix	I	A/R	C	C	I
Nominate data owners and focal points	C	A/R	I	C	I
Classify priority datasets	C	R	A/R	C	I
Complete metadata and data dictionary records	I	C	A/R	C	I
Apply QA/QC and reliability grading	I	C	A/R	C	I
Approve public release	C	C	A/R	C	I
Approve restricted or confidential access	C	C	R	A/R	I
Operate portal and integration services	I	A/R	C	C	I
Report KPI progress	A	R	C	C	I

Chapter 5. Implementation Workstreams

The five workstreams translate the five approved NEDMP pillars into integrated implementation packages.

Legend: WS means workstream. WS1-WS5 correspond to the five NEDMP implementation workstreams that operationalize the five foundational pillars.

WS1. Data Governance and Institutional Management

Outcome: A nationally coordinated, role-based environmental data governance system. Evidence base: S1 p. 17; S6 p. 29; S4II pp. 63-68.

- National governance framework approval
- Institutional role clarification and decision rights
- Focal-point network and data stewardship model
- Data-sharing agreements and compliance procedures
- SOP development and governance maturity reviews

WS2. Data Quality

Outcome: Priority environmental data are traceable, reliable, comparable and fit for assessment and reporting. Evidence base: S1 pp. 18-19; S6 pp. 31-32; S4III p. 11.

- National dataset inventory and registry
- Metadata framework and data dictionary
- QA/QC dimensions, validation and reliability grading
- Indicator catalogue and source documentation
- Data quality improvement plans

WS3. Data Interoperability and System Integration

Outcome: Environmental data exchange shifts from ad hoc requests to secure recurrent flows and system-to-system integration. Evidence base: S1 pp. 19-20; S4III pp. 10-12; S6 pp. 35-38.

- Systems inventory and readiness assessment
- Enterprise/data architecture and exchange protocols
- Standard formats, controlled vocabularies and APIs
- eMISK integration pathway
- Data submission and integration testing

WS4. Data Sharing, Accessibility and Transparency

Outcome: Public access expands safely through classification, approval, open-data readiness and controlled-release rules. Evidence base: S1 p. 20; S6 pp. 27-28; S4III pp. 9-11.

- Access policy and request workflow
- Public/internal/restricted/confidential classification
- Publication-readiness checklist
- Open-data pathway and knowledge products
- Data access monitoring and feedback

WS5. Capacity Development and Sustainability

Outcome: Institutions have the skills, staffing, financing and continuity arrangements needed to sustain NEDMP. Evidence base: S1 p. 21; S4II pp. 71-73; S6 pp. 39-41.

- Competency framework and training plan
- Data steward and focal-point certification
- Sustainable financing and annual workplanning
- Cybersecurity, backup and business-continuity controls
- Continuous improvement and adaptive review

WS1 Governance	WS2 Quality	WS3 Interoperability	WS4 Sharing	WS5 Capacity
Roles, rules, decision rights	Registry, metadata, QA/QC	Standards, APIs, eMISK links	Access model, open data	Skills, finance, sustainability

Figure 8. Five-workstream implementation model

Legend: Programme codes use the format P1.1-P5.4. P means Programme; the first number maps to the related workstream/pillar, and the decimal number identifies the sequence within that workstream. The Workstream column uses WS1-WS5.

Table 7. Workstream programme matrix

Programme code	Programme	Workstream code	Timeline	Expected output	Source evidence
P1.1	Governance activation	WS1	Immediate	Governance structure operational	S6 p. 29
P1.2	Ownership and stewardship	WS1	Immediate to short term	Role register and ownership records	S6 p. 29
P1.3	SOP and agreement package	WS1	Short term	Approved SOP library	S1 p. 27; S4II p. 72
P2.1	Dataset registry	WS2	Immediate	National dataset registry	S6 p. 31
P2.2	Metadata and quality standards	WS2	Immediate to short term	Minimum quality standard package	S6 pp. 31-32
P2.3	Indicator catalogue	WS2	Short term	Environmental indicator catalogue	S4II p. 72
P3.1	Systems readiness assessment	WS3	Immediate	Systems and readiness baseline	S6 pp. 13-14
P3.2	Exchange protocols and API-readiness	WS3	Short to medium term	Interoperability standard package	S4III p. 10; S6 p. 13
P3.3	eMISK-linked integration pilots	WS3	Medium term	Validated exchange pilots	S4II p. 71; S6 p. 38
P4.1	Classification and access model	WS4	Immediate	Classification policy and workflow	S6 pp. 27-28
P4.2	Open-data readiness pathway	WS4	Short to medium term	Phased open-data releases	S4II p. 71; S6 p. 16
P4.3	Access request workflow	WS4	Short term	Tracked access decisions	S6 p. 31
P5.1	Competency framework	WS5	Immediate	Competency framework	S4II p. 72
P5.2	Annual capacity programme	WS5	Continuous	Training completion records	S1 p. 21; S6 p. 30
P5.3	Sustainable financing	WS5	Immediate to medium term	Indicative resource plan	S6 pp. 14, 39
P5.4	Continuity and cyber resilience	WS5	Short to medium term	Continuity control plan	S6 pp. 13, 41

Chapter 6. Implementation Programmes, Projects and Sequencing

The programme structure turns the workstreams into sequenced outputs, while leaving activity-level planning to the future Action Plan.

6.1 Five-year implementation phases

Table 8. Five-year implementation phase matrix

Phase	Title	Indicative timing	Main activities	Primary outputs
Phase 1	Institutional endorsement and mobilization	0-6 months	Approve implementation governance, focal points, evidence baseline and workplan.	Mandate, governance bodies and PMO activated.
Phase 2	Rules, inventory and standards foundation	6-18 months	Inventory datasets, assign owners, issue classification, metadata and QA/QC templates.	Registered priority datasets and minimum standards package.
Phase 3	Priority implementation and early integration	18-30 months	Pilot data flows, classification, access workflows, eMISK-linked exchange and open-data releases.	Validated pilots, quality gates and first publication pathway.
Phase 4	Scale-up and institutionalization	30-48 months	Extend to additional institutions, automate recurrent exchange and embed annual budgets.	Operational national environmental data governance system.
Phase 5	Evaluation, adaptation and next cycle	48-60 months	Conduct final review, update standards, confirm sustainability and prepare next NEDMP cycle.	Evaluated and updated implementation model.

Year 1	Year 2	Year 3	Year 4	Year 5
Mobilize governance; approve rules	Registry, standards and classification	Integration pilots and publication pathway	Scale institutional coverage and portal services	Final evaluation, update and next cycle

Figure 9. Five-year phased roadmap

6.2 Quick wins

Table 9. Quick-wins matrix

Quick win	Why it is feasible	Dependency	Expected value
Focal-point registry	Government institutions already form the main respondent base.	Nomination request	Clear coordination channel.
Priority dataset inventory	Data are already produced regularly.	Inventory template	Known owners, sources and gaps.
Metadata template	Source reports identify uneven documentation.	Approval of minimum fields	Immediate quality improvement.
Classification policy draft	S6 provides classification levels and criteria.	Legal/security review	Controlled sharing and open-data pathway.
Pilot publication of low-risk indicators	Some datasets are already suitable for public information.	QA/QC and classification	Visible early transparency gain.

Foundation	Quality gate	Integration gate	Access gate	Sustainability gate
Governance roles	Metadata and QA/QC	Standard formats and APIs	Classification and approval	Training, budget and review

Figure 10. Workstream dependency diagram

Chapter 7. Data Standards, Classification and Portal Integration

This chapter translates the data classification framework and portal requirements into implementation controls.

7.1 Environmental data classification

Classification is the operational control that allows Kuwait to increase transparency while protecting sensitive, restricted and legally protected data. The model should be applied to priority datasets before publication or platform integration (NEDMP Data Classification Governance Framework, p. 27; NEDMP Data Classification Governance Framework, p. 28).

Table 10. Environmental data classification and access matrix

Level	Classification	Typical data	Default access rule
Level 1	Public / Open Data	Validated, non-sensitive and legally releasable environmental indicators or datasets.	Open publication with metadata and QA/QC status.
Level 2	Controlled Public Release	Aggregated or summarized data requiring context, limitation notes or delayed release.	Publish with controls, aggregation or explanatory notes.
Level 3	Government Internal	Raw, draft, operational or non-validated datasets for official use.	Authorized government users and focal points.
Level 4	Restricted / Sensitive	Facility-level, compliance, security, sensitive habitat, commercial or institutional-risk data.	Purpose-based approval, named users and access logs.
Level 5	Confidential / Legally Protected	Legally restricted, contractual, personal, security-sensitive or high-risk data.	Formal legal/security authorization only.

Dataset registered	Quality checked	Classification decision	Legal/security review	Release or controlled access
Owner, steward, source, theme and update cycle recorded	Metadata, QA/QC and limitation status verified	Public, controlled, internal, restricted or confidential	Triggered for sensitive or legally constrained data	Open publication, internal sharing or authorized release

Figure 11. Data classification and access workflow

7.2 Portal integration

The National Environmental Data Portal should be treated as a governance service, not as a stand-alone IT project. It should operationalize the registry, metadata, classification, access request, publication and reporting functions that the NEDMP governance system approves.

Table 11. National Environmental Data Portal functional requirements

Portal module	Main functions	Expected governance value
Institutional onboarding and focal point registry	Register institutions, roles, permissions and contact points.	Creates accountability and coordination.
Dataset registry and metadata catalogue	Record dataset descriptions, owners, stewards, methods, update frequency and quality status.	Improves discoverability and traceability.
Classification and access-control engine	Apply public, internal, restricted, sensitive and confidential levels.	Controls publication and access decisions.
Data submission and validation module	Receive datasets, check formats, metadata and QA/QC status.	Creates quality gates before integration.

Interoperability/API services	Support recurrent exchange with approved formats, identifiers and API-readiness criteria.	Reduces request-based exchange.
Open-data and knowledge product interface	Publish approved indicators, datasets, dashboards and reports.	Improves transparency and reuse.
Analytics and reporting interface	Generate dashboards, gaps analysis, reports and progress indicators.	Supports decision-making and monitoring and evaluation (M&E).

Institutional systems	Registry and metadata	Quality and classification gates	Access and publication services	Users and reporting
eMISK, repositories, GIS, monitoring systems	Dataset catalogue, owners, stewards, data dictionary	QA/QC, limitations, classification, approval	APIs, dashboards, data requests, open data	Decision-makers, agencies, researchers, public, MEA/SDG reporting

Figure 12. National Environmental Data Portal architecture

Producer	Owner/steward	Registry	QA/QC	Portal/API	User
Collects or submits data	Approves metadata, classification and quality	Records dataset and update schedule	Validates, grades and logs limitations	Publishes or shares according to access class	Uses data for policy, reporting or public information

Figure 13. Environmental data flow from producer to user

Register	Document	Validate	Classify	Approve	Publish/share
Dataset owner and steward identified	Metadata, methods and source recorded	QA/QC and reliability grading completed	Access class assigned	Owner/legal/security approval as needed	Portal publication or controlled exchange

Figure 14. Dataset onboarding and quality approval process

7.3 Standards and interoperability alignment

International standards and good practices should be applied as implementation aids. They should not be presented as new national obligations unless adopted by competent authorities. Their practical role is to inform metadata, statistics, spatial data, quality, exchange, security, continuity and risk controls.

Table 12. Standards and interoperability matrix

Framework or standard	Implementation use	Application in NEDMP	Reference
UNEP Global Environmental Data Strategy	National environmental data ecosystem, trust, inclusion, accessibility and interoperability	Use as the overarching environmental data governance reference; current UNEP materials describe GEDS as a voluntary enabling framework and not a new reporting obligation.	UNEP GEDS official page; S2 pp. 11-12
FDES 2013	Environment statistics domains, components and basic statistical structure	Use to organize environmental statistics and indicator catalogue.	UN Statistics Division, 2013; S3 p. 12
SEEA	Environmental-economic accounts and integrated environment/economy datasets	Use where environmental data support accounting and SDG indicators.	UN et al., 2012; S3 p. 12
SDMX	Statistical data and metadata exchange	Use for structured statistical reporting and machine-readable exchange; SDMX 3.1 technical specifications were released in May 2025.	SDMX.org technical specifications

FAIR principles	Findability, accessibility, interoperability and reuse	Use in metadata, registry and publication-readiness checks.	Wilkinson et al., 2016; S3 p. 23
SEIS principles	Collect once, share often, public access and interoperability	Use in data sharing and portal services.	S3 pp. 15-16
INSPIRE and spatial data infrastructure practice	Spatial metadata, services and interoperability	Use for GIS and geospatial exchange requirements.	European Commission INSPIRE
ISO 19115-1:2014	Geographic metadata	Use for spatial dataset metadata profile.	ISO 19115-1:2014
ISO 8000 series	Data quality and master data principles	Use as reference for data quality management principles.	ISO 8000 series
ISO/IEC 27001:2022	Information security management systems	Use for security governance and access control alignment.	ISO/IEC 27001:2022
ISO 22301:2019	Business continuity management systems	Use for continuity and disaster recovery controls.	ISO 22301:2019
ISO 31000:2018	Risk management guidelines	Use for risk register, ownership and review cycle.	ISO 31000:2018
DAMA-DMBOK	Data management knowledge areas	Use as a general data management practice reference, not as a legal mandate.	DAMA International

Governance	Statistics	Spatial data	Exchange	Security and resilience
UNEP GEDS, SEIS, DAMA-DMBOK	FDES, SEEA, SDG indicators	INSPIRE, ISO 19115-1	SDMX, APIs, metadata profiles	ISO/IEC 27001, ISO 22301, ISO 31000

Figure 15. Standards-to-implementation alignment

Chapter 8. Resource Mobilization and Financing Framework

The Implementation Plan defines resource categories and financing logic without inventing unapproved budget amounts.

8.1 Resource principles

Resource planning should distinguish initial set-up costs from recurrent operating costs. Sustainable financing must cover coordination, metadata management, QA/QC, platform operation, capacity development, cybersecurity, monitoring and periodic review (Section I - National Vision, Objectives and Pillars of Environmental Data Governance, p. 8; NEDMP Data Classification Governance Framework, p. 14).

Table 13. Indicative resource requirements matrix

Resource area	Indicative requirements	Timing	Possible financing channel	Budget note
Governance and PMO	Secretariat staff, committee support, legal/compliance support, implementation reporting	Immediate and recurrent	Core government budget; programme support	No exact amount proposed without costing approval.
Standards and dataset registry	Metadata specialists, data stewards, registry platform, quality templates	Immediate to short term	Core budget; technical assistance	Costing to follow dataset prioritization.
Portal and interoperability	Architecture design, platform modules, API development, security, hosting and integration pilots	Short to medium term	Capital investment plus operating budget	Phased to avoid technology preceding governance.
Capacity development	Training materials, certification, workshops, coaching and helpdesk	Continuous	Annual training budget; partner support	Targeted to focal points, stewards, GIS/database and reporting teams.
Monitoring and evaluation	KPI data collection, annual review, mid-term review and final evaluation	Annual and milestone-based	Programme management budget	Evaluation costs should be built into annual workplans.

National budget planning	NEDMP annual workplan	Approved programme resources	Implementation delivery	Performance review
Annual resource needs are identified	Workstreams and outputs are prioritized	Budget lines and partner support are confirmed	Funds support governance, standards, portal, capacity and monitoring and evaluation (M&E)	Results inform next budget and workplan

Figure 16. Financing and resource flow

Chapter 9. Risk Management and Compliance

Risk management should be integrated into governance, quality, access and platform decisions from the start.

Table 14. NEDMP implementation risk register

Risk	Description	Likelihood	Impact	Primary mitigation	Source
Governance and coordination risk	Fragmented accountability delays exchange and conflict resolution.	High	High	National governance committee, focal-point network, data owner/steward roles and escalation matrix.	S6 pp. 12-14
Access and transparency risk	Institutions may over-restrict data or release it without safeguards.	High	High	Public/internal/restricted/confidential rules, publication checklists and legal review.	S6 pp. 27-28
Quality and metadata risk	Datasets may not be traceable, comparable or reliable.	High	High	Mandatory metadata, QA/QC, reliability grading and audit trails.	S6 pp. 31-32
Interoperability risk	Data flows remain manual and inconsistent.	Medium	High	Standard formats, controlled vocabularies, exchange protocols and API-readiness criteria.	S4III p. 10; S6 p. 13
Systems and resilience risk	Platform integration exposes cybersecurity, backup or recovery gaps.	Medium	High	Systems readiness baseline, DR plan, cybersecurity controls and phased integration.	S6 pp. 13, 41
Capacity risk	Implementation burden exceeds available skills and staffing.	High	Medium	Competency framework, role-based training and steward certification.	S4II p. 72; S6 p. 30
Financing risk	Classification, metadata, QA/QC and platform operation are not sustained.	Medium	High	Annual budget lines, phased resource plan and management review.	S6 p. 14
Legal non-compliance risk	Sensitive or legally restricted data are shared inappropriately.	Medium	High	Legal/security clearance for restricted/confidential access and release records.	S6 p. 32

Lower impact	Medium impact	High impact
Monitor: stakeholder engagement delays	Mitigate: capacity and financing risk	Escalate: governance, legal, quality and access risk
Monitor: minor reporting delay	Mitigate: systems readiness risk	Escalate: cyber/continuity risk

Figure 17. Risk heatmap

9.1 Compliance controls

Compliance controls include legal review for restricted and confidential data, documented release decisions, access logs, audit trails, backup and disaster recovery arrangements, and cybersecurity safeguards. These controls are prerequisites for trusted sharing and public access.

Chapter 10. Monitoring, Evaluation and Learning Framework

The monitoring framework uses implementation KPIs to connect outputs, outcomes, responsible roles and means of verification.

Legend: KPI means key performance indicator. SO means strategic objective; SO1-SO5 are the five NEDMP strategic objectives. Pillar refers to the NEDMP foundational pillars.

Table 15. Results and KPI framework

KPI area	Indicator	Baseline	Target	Frequency	Lead	Means of verification	Objective/pillar
Governance activation	NEDMP governance bodies operational	Not yet formalized	Steering and governance committees operating with terms of reference (TORs)	Quarterly	EPA/KEPA secretariat	Decision records; terms of reference (TORs)	SO1; Pillar 1
Institutional coverage	Participating institutions with nominated focal points	To be confirmed at mobilization	All priority institutions	Quarterly	EPA/KEPA secretariat	Focal-point registry	SO1; Pillar 1
Dataset registration	Priority datasets registered with owner, steward and custodian	No consolidated national registry	100 per cent of priority datasets	Quarterly	Dataset registry function	Registry records	SO2; Pillar 2
Metadata completeness	Registered priority datasets with mandatory metadata complete	Uneven metadata documentation	At least 90 per cent by end of Phase 3	Quarterly	Data stewards	Metadata audit	SO2; Pillar 2
QA/QC compliance	Priority datasets passing minimum QA/QC before release or integration	Mixed QA/QC practice	At least 85 per cent by end of Phase 3	Quarterly	QA/QC focal points	Validation logs	SO2; Pillar 2
Interoperability readiness	Priority recurrent data flows using standard formats/API-ready specifications	Partial readiness	At least 10 pilot flows by Phase 3; scale thereafter	Semi-annual	Technical implementation unit	Integration tests	SO3; Pillar 3
Classification coverage	Priority datasets with approved classification level	No uniform national model	100 per cent of priority datasets	Quarterly	Data owners	Classification register	SO4; Pillar 4
Open-data pathway	Validated non-sensitive datasets or indicators published	Full public availability 18.8 per cent	Annual increase in publishable datasets	Semi-annual	Portal operating function	Publication log	SO4; Pillar 4
Capacity development	Focal points, stewards and technical staff completing role-based training	Uneven training availability	Annual training plan implemented	Annual	Capacity-development lead	Training records	SO5; Pillar 5
Financing and sustainability	Annual workplan includes NEDMP operating and maintenance resources	Funding divergence reported	Budget lines proposed and reviewed annually	Annual	EPA/KEPA with finance functions	Budget/workplan records	SO5; Pillar 5
Continuity and security	Priority systems with backup, disaster recovery and cybersecurity controls documented	Uneven resilience readiness	All priority repositories by Phase 4	Semi-annual	Cybersecurity and custodians	DR/security audit	SO3; SO5

Adaptive review	Annual implementation review completed and actioned	New mechanism	Annual review and management response	Annual	Steering Committee	Review report	All SOs

Collect evidence	Validate progress	Review risks	Report decisions	Adjust plan
Focal points and workstreams submit KPI evidence	Secretariat checks records and sources	Risk register and dependencies updated	Committee reviews and issues decisions	Annual workplan and standards updated

Figure 18. Monitoring and reporting cycle

10.1 Evaluation moments

- Annual implementation review.
- Mid-term implementation review at the end of Phase 3.
- Final evaluation before the next NEDMP implementation cycle.
- Targeted technical reviews when standards, laws, platforms or reporting obligations change.

Chapter 11. Implementation Schedule and Critical Path

The schedule is structured around dependencies: governance activates standards, standards enable classification and quality controls, and those controls enable integration and publication.

Table 16. Dependency and critical-path matrix

Dependency	Must be completed before	Risk if delayed	Management response
Governance bodies and focal points	Dataset registry, classification approvals and integration pilots	No clear decision rights	Issue terms of reference (TORs) and focal-point nomination request early.
Dataset inventory	Metadata completion, classification and gap closure	Unknown datasets and owners	Use a simple template and prioritize high-value datasets.
Metadata and QA/QC rules	Public release and platform integration	Low trust and inconsistent data	Approve minimum mandatory fields and validation rules.
Classification rules	Open-data pathway and controlled access	Over-release or over-restriction	Apply legal/security review triggers.
Systems readiness baseline	API/integration design	Technology misfit	Assess repositories, eMISK and institutional readiness first.

Chapter 12. Capacity Development and Change Management

Capacity development is a core implementation workstream, not a support activity. It should be role-based, repeated and connected to actual dataset onboarding and reporting tasks.

Table 17. Capacity-development matrix

Target group	Priority competencies	Delivery approach	Expected result
Senior decision-makers	Mandate, governance decisions, data sovereignty, risk ownership	Executive briefings and annual review sessions	Steering decisions reflect evidence and risk.
Institutional focal points	Coordination, reporting, dataset registration, issue escalation	Induction, quarterly clinics and peer network	Active focal-point network.
Data stewards	Metadata, QA/QC, classification, versioning, source documentation	Certification modules and supervised dataset onboarding	Priority datasets meet minimum quality controls.
GIS and database teams	Spatial metadata, APIs, interoperability, database management and security	Technical workshops and integration pilots	Reusable integration patterns.
Legal/compliance teams	Access rules, agreements, confidentiality and publication decisions	Legal workflow workshops and template review	Traceable release decisions.
Reporting and assessment teams	Indicator catalogue, SOE reporting, SDG/MEA requirements and data interpretation	Reporting clinics and indicator catalogue training	Improved reporting readiness.

Awareness	Role induction	Technical modules	Applied onboarding	Certification and refresh
Senior briefings and institutional ownership	Focal-point and steward roles	Metadata, QA/QC, GIS, APIs, legal access	Dataset registration and pilot integration	Annual refresh and competency records

Figure 19. Capacity-development pathway

Chapter 13. Stakeholder Engagement and Communication

Stakeholder engagement should maintain national ownership, manage expectations, collect feedback and support data providers and users.

Table 18. Stakeholder engagement matrix

Stakeholder group	Influence/interest	Engagement mechanism	Frequency
Government ministries and agencies	High influence / high interest	Formal committee participation, focal-point network and data-sharing agreements	Quarterly
EPA/KEPA departments	High influence / high interest	Implementation unit, thematic groups, eMISK and dataset onboarding	Monthly during start-up
Central Statistical Bureau and planning entities	High influence / medium interest	Indicator alignment, official statistics and national planning coordination	Quarterly
Research and academic institutions	Medium influence / high interest	Technical working groups, data standards and research data contribution	Semi-annual
Private sector and regulated entities	Medium influence / medium interest	Controlled data submission rules, compliance guidance and confidentiality safeguards	Semi-annual
Civil society and public users	Low influence / high interest	Public information products, open data releases and feedback channel	Annual plus portal feedback
International and regional partners	Medium influence / medium interest	Technical assistance, alignment review and capacity support	As needed

13.1 Communication products

- Annual NEDMP implementation progress note.
- Dataset registry and indicator catalogue release notes.
- Open-data publication announcements.
- Technical guidance notes for data owners and stewards.
- Public-facing environmental data explainers for released datasets.

Chapter 14. Sustainability and Institutionalization

Sustainability requires legal, institutional, financial, technical, human resource and knowledge continuity.

Table 19. Sustainability matrix

Sustainability dimension	Implementation requirement	Institutionalization mechanism
Legal	Access, classification and sharing rules align with national law.	Legal review workflow and approved templates.
Institutional	Governance bodies and focal points continue beyond project start-up.	Terms of reference and annual reporting cycle.
Financial	Recurrent costs are visible in annual workplans.	Budget line proposals and phased financing framework.
Technical	Portal, eMISK links, APIs, backups and cybersecurity are maintained.	Service ownership and maintenance plans.
Human resources	Data stewardship skills are retained and refreshed.	Competency framework and training records.
Knowledge	Standards, templates and evidence are documented.	NEDMP knowledge repository and periodic review.

Legal	Institutional	Financial	Technical	Human	Knowledge
Rules and mandates	Committees and roles	Budget and resources	Portal and security	Skills and staffing	Templates and evidence

Figure 20. NEDMP sustainability model

Chapter 15. Review, Adaptive Management and Next Steps

The NEDMP should be implemented as a living national instrument that can respond to institutional capacity, technology, reporting obligations and environmental priorities.

15.1 Review cycle

Annual reviews should assess delivery progress, KPI evidence, risk status, institutional participation, source-data gaps, standards implementation, capacity needs and budget implications. A mid-term review should test whether the implementation sequence remains appropriate. A final evaluation should inform the next NEDMP cycle.

Plan	Implement	Monitor	Review	Adapt
Annual workplan and standards	Workstreams and programmes	KPIs, risks and evidence	Committee and stakeholder review	Updated priorities, rules and next action plan

Figure 21. Adaptive-management feedback loop

15.2 Immediate next steps

- Validate this Implementation Plan with EPA/KEPA and participating institutions.
- Confirm proposed governance bodies, roles and approval pathways through national processes.
- Prepare the first annual NEDMP Action Plan with named activities, deadlines and approved resources.
- Launch the focal-point network and priority dataset inventory.
- Approve minimum classification, metadata and QA/QC templates before platform integration is scaled.

Annexes

Annex A. Detailed implementation results framework

Result level	Statement	Indicator
Impact	Environmental governance is supported by trusted data.	Use of NEDMP outputs in policy/reporting
Outcome	Governance, quality, interoperability, access and capacity improve.	KPI framework
Output	Rules, registries, templates and portal modules delivered.	Output completion records

Annex B. Five-year implementation matrix

Phase	Main output	Lead
Phase 1	Mandate, governance bodies and PMO activated.	EPA/KEPA secretariat with approved governance bodies
Phase 2	Registered priority datasets and minimum standards package.	EPA/KEPA secretariat with approved governance bodies
Phase 3	Validated pilots, quality gates and first publication pathway.	EPA/KEPA secretariat with approved governance bodies
Phase 4	Operational national environmental data governance system.	EPA/KEPA secretariat with approved governance bodies
Phase 5	Evaluated and updated implementation model.	EPA/KEPA secretariat with approved governance bodies

Annex C. Institutional roles and responsibilities

Role	Status	Function
High-level national oversight mechanism	Proposed, subject to approval	Endorse priorities, resolve inter-ministerial issues and connect NEDMP to national planning.
NEDMP Steering Committee	Proposed, subject to approval	Provide strategic direction, approve annual workplans and review implementation performance.
National Environmental Data Governance Committee	Proposed, subject to approval	Approve standards, classification rules, dataset prioritization and data-sharing protocols.
EPA/KEPA secretariat and coordination function	Existing mandate plus proposed implementation function	Coordinate implementation, maintain the registry, prepare reporting and support committees.
Programme management office or technical implementation unit	Proposed, subject to approval	Manage delivery schedule, dependencies, risks, evidence matrix and progress reporting.
Thematic technical working groups	Proposed	Develop domain-specific metadata, QA/QC, indicators and integration requirements.
Institutional environmental data focal points	Proposed network	Coordinate within institutions, maintain dataset records and facilitate exchange.
Data owners	Existing responsibilities to be formalized	Approve classification, correction, sharing and publication decisions for assigned datasets.

Annex D. RACI matrix

Activity	Steering	EPA/KEPA	Data owner/steward	Legal/security	Public

Approve NEDMP implementation governance	A	R	C	C	I
Maintain source-evidence matrix	I	A/R	C	C	I
Nominate data owners and focal points	C	A/R	I	C	I
Classify priority datasets	C	R	A/R	C	I
Complete metadata and data dictionary records	I	C	A/R	C	I
Apply QA/QC and reliability grading	I	C	A/R	C	I
Approve public release	C	C	A/R	C	I
Approve restricted or confidential access	C	C	R	A/R	I

Annex E. Governance committee terms of reference

Element	Indicative content	Approval note
Mandate	Approve standards, classification rules and cross-institutional governance decisions.	Subject to national approval
Membership	EPA/KEPA, key data-producing institutions, legal/security and statistical/planning representatives.	To be confirmed
Meeting frequency	Quarterly, with extraordinary meetings for escalation.	Indicative

Annex F. Technical working group terms of reference

Group	Indicative focus	Deliverable
Data quality and metadata	Metadata, QA/QC, data dictionary and reliability grading.	Minimum standards package
Interoperability and portal	Architecture, formats, APIs and eMISK integration.	Integration roadmap
Access and classification	Classification, open-data readiness and release workflow.	Access protocol

Annex G. NEDMP programme management structure

Function	Responsibility	Output
Secretariat	Coordinate meetings, reporting and evidence matrix.	Quarterly progress report
PMO	Manage dependencies, schedule and risks.	Implementation dashboard
Workstream leads	Coordinate workstream outputs.	Workstream deliverables

Annex H. Indicative Gantt chart

Work package	Year 1	Year 2	Year 3	Year 4	Year 5
Governance activation	Start/complete	Operate	Operate	Operate	Evaluate

Dataset registry	Start	Scale	Maintain	Maintain	Update
Portal/integration	Prepare	Pilot	Scale	Operate	Review
Capacity	Start	Continue	Continue	Continue	Refresh

Annex I. Dependency and critical-path matrix

Dependency	Before	Response
Governance roles	Classification and sharing	Approve terms of reference (TORs) and nominations
Metadata/QA/QC	Publication and integration	Approve minimum standard
Systems readiness	Portal integration	Conduct readiness assessment

Annex J. Risk register

Risk	Level	Mitigation
Governance and coordination risk	High likelihood / High impact	National governance committee, focal-point network, data owner/steward roles and escalation matrix.
Access and transparency risk	High likelihood / High impact	Public/internal/restricted/confidential rules, publication checklists and legal review.
Quality and metadata risk	High likelihood / High impact	Mandatory metadata, QA/QC, reliability grading and audit trails.
Interoperability risk	Medium likelihood / High impact	Standard formats, controlled vocabularies, exchange protocols and API-readiness criteria.
Systems and resilience risk	Medium likelihood / High impact	Systems readiness baseline, DR plan, cybersecurity controls and phased integration.
Capacity risk	High likelihood / Medium impact	Competency framework, role-based training and steward certification.
Financing risk	Medium likelihood / High impact	Annual budget lines, phased resource plan and management review.
Legal non-compliance risk	Medium likelihood / High impact	Legal/security clearance for restricted/confidential access and release records.

Annex K. KPI metadata sheets

KPI	Definition	Means of verification
Governance activation	NEDMP governance bodies operational	Decision records; terms of reference (TORs)
Institutional coverage	Participating institutions with nominated focal points	Focal-point registry
Dataset registration	Priority datasets registered with owner, steward and custodian	Registry records
Metadata completeness	Registered priority datasets with mandatory metadata complete	Metadata audit
QA/QC compliance	Priority datasets passing minimum QA/QC before release or integration	Validation logs
Interoperability readiness	Priority recurrent data flows using standard formats/API-ready specifications	Integration tests
Classification coverage	Priority datasets with approved classification level	Classification register

Open-data pathway	Validated non-sensitive datasets or indicators published	Publication log
Capacity development	Focal points, stewards and technical staff completing role-based training	Training records
Financing and sustainability	Annual workplan includes NEDMP operating and maintenance resources	Budget/workplan records
Continuity and security	Priority systems with backup, disaster recovery and cybersecurity controls documented	DR/security audit
Adaptive review	Annual implementation review completed and actioned	Review report

Annex L. Monitoring and reporting templates

Template	Minimum fields	Frequency
Quarterly progress report	Output status, KPI evidence, risks, decisions needed	Quarterly
Annual review	Results, budget, lessons, corrective actions	Annual

Annex M. National environmental dataset registry template

Field	Requirement	Notes
Dataset title	Mandatory	Clear title and description
Owner/steward/custodian	Mandatory	Named roles
Update frequency	Mandatory	Expected recurrence
Classification	Mandatory	Level 1-5

Annex N. Minimum metadata template

Field	Requirement	Notes
Theme and sub-theme	Mandatory	Use approved environmental themes
Source method	Mandatory	Monitoring, administrative, GIS, survey etc.
Limitations	Mandatory	Known gaps and caveats

Annex O. Data quality assessment template

Dimension	Check	Evidence
Accuracy	Values checked against source/method.	Validation log
Completeness	Missing values and gaps documented.	Gap record
Timeliness	Update delay within threshold.	Timestamp audit

Annex P. Dataset classification assessment template

Criterion	Question	Decision implication
Public value	Does it support transparency/reporting?	Consider public release
Sensitivity	Could release create risk?	Restrict or aggregate

Legal restriction	Is legal review required?	Restricted/confidential

Annex Q. Data-sharing agreement template

Clause	Indicative content	Owner
Purpose	Define approved use and dataset scope.	Data owner
Access conditions	Users, duration, safeguards and logs.	Legal/security
Quality and limitations	Known limitations and update cycle.	Data steward

Annex R. Data access request workflow

Step	Action	Responsible
1	Submit request and purpose.	Requester
2	Check classification and metadata.	Portal/operator
3	Owner/legal/security review if triggered.	Owner and legal/security
4	Release, reject or request revision.	Approving authority

Annex S. SOP development schedule

SOP	Priority	Timing
Dataset registration	High	Phase 1
Metadata completion	High	Phase 1-2
QA/QC validation	High	Phase 2
Publication workflow	High	Phase 2-3

Annex T. Capacity-development matrix

Target group	Competency	Delivery
Senior decision-makers	Mandate, governance decisions, data sovereignty, risk ownership	Executive briefings and annual review sessions
Institutional focal points	Coordination, reporting, dataset registration, issue escalation	Induction, quarterly clinics and peer network
Data stewards	Metadata, QA/QC, classification, versioning, source documentation	Certification modules and supervised dataset onboarding
GIS and database teams	Spatial metadata, APIs, interoperability, database management and security	Technical workshops and integration pilots
Legal/compliance teams	Access rules, agreements, confidentiality and publication decisions	Legal workflow workshops and template review
Reporting and assessment teams	Indicator catalogue, SOE reporting, SDG/MEA requirements and data interpretation	Reporting clinics and indicator catalogue training

Annex U. Indicative resource and budget template

Resource area	Cost category	Approval note

Governance and PMO	Secretariat staff, committee support, legal/compliance support, implementation reporting	Exact figures require national costing and approval.
Standards and dataset registry	Metadata specialists, data stewards, registry platform, quality templates	Exact figures require national costing and approval.
Portal and interoperability	Architecture design, platform modules, API development, security, hosting and integration pilots	Exact figures require national costing and approval.
Capacity development	Training materials, certification, workshops, coaching and helpdesk	Exact figures require national costing and approval.
Monitoring and evaluation	KPI data collection, annual review, mid-term review and final evaluation	Exact figures require national costing and approval.

Annex V. Communication and stakeholder engagement matrix

Stakeholder	Mechanism	Frequency
Government ministries and agencies	Formal committee participation, focal-point network and data-sharing agreements	Quarterly
EPA/KEPA departments	Implementation unit, thematic groups, eMISK and dataset onboarding	Monthly during start-up
Central Statistical Bureau and planning entities	Indicator alignment, official statistics and national planning coordination	Quarterly
Research and academic institutions	Technical working groups, data standards and research data contribution	Semi-annual
Private sector and regulated entities	Controlled data submission rules, compliance guidance and confidentiality safeguards	Semi-annual
Civil society and public users	Public information products, open data releases and feedback channel	Annual plus portal feedback
International and regional partners	Technical assistance, alignment review and capacity support	As needed

Annex W. Source-evidence traceability matrix

Intervention	Supporting finding	Source	Strategic objective code	Pillar	Output
Governance-first sequencing	Governance readiness is lower than technical readiness.	S4III p. 5; S6 p. 7	SO1	Pillar 1	Governance structures and decision rights activated first.
Dataset registry	Data availability is substantial but gaps, sources and update cycles vary.	S4III pp. 5, 9; S6 p. 21	SO2	Pillar 2	Priority dataset registry and indicator catalogue.
Classification model	Public access is limited and data sensitivity/legal restrictions constrain sharing.	S6 pp. 16, 27-28, 32	SO4	Pillar 4	Public/internal/restricted/confidential model.
QA/QC and metadata	QA/QC and metadata practices are uneven across institutions.	S4III p. 11; S6 pp. 31-32	SO2	Pillar 2	Minimum metadata, QA/QC and reliability grading.
Interoperability roadmap	Exchange is active but often request-based and only partially standardized.	S4III pp. 10-11; S6 p. 13	SO3	Pillar 3	Standard formats, API-readiness and integration pilots.
eMISK and portal integration	National portal and eMISK are identified as priority outputs.	S4II p. 71; S6 pp. 35-38	SO3; SO4	Pillars 3 and 4	Portal modules and phased platform integration.

Capacity programme	Skills, staffing and training are visible gaps.	S4II p. 72; S6 p. 30	SO5	Pillar 5	Role-based competency and training programme.
Financing model	Funding divergence threatens sustainability.	S4II p. 73; S6 p. 14	SO5	Pillar 5	Annual resource plan and budget lines.
Risk controls	Cybersecurity, continuity, access and legal risks require controls.	S6 pp. 13, 32, 41	SO1; SO3; SO4	Pillars 1, 3 and 4	Risk register and continuity/security controls.
Monitoring framework	Implementation indicators are required to track progress.	NEDMP p. 30; S6 p. 39	All SOs	All pillars	KPI framework, annual review and adaptive management.

Annex X. Glossary and references

Term	Definition	Reference
Environmental data governance	Decision rights, roles, rules and controls for environmental data.	NEDMP sources
Data steward	Operational role maintaining metadata, quality and dataset status.	Implementation Plan
Open data	Validated, non-sensitive and legally releasable data published for reuse.	Classification matrix

References

DAMA International (2017). DAMA-DMBOK: Data Management Body of Knowledge, 2nd edition. Technics Publications.

European Commission (2007). Directive 2007/2/EC establishing an Infrastructure for Spatial Information in the European Community (INSPIRE).

ISO (2014, amended 2018 and 2020). ISO 19115-1:2014 Geographic information - Metadata - Part 1: Fundamentals.

ISO (2018). ISO 31000:2018 Risk management - Guidelines.

ISO (2019). ISO 22301:2019 Security and resilience - Business continuity management systems - Requirements.

ISO/IEC (2022, amended 2024). ISO/IEC 27001:2022 Information security, cybersecurity and privacy protection - Information security management systems - Requirements.

ISO (various). ISO 8000 series on data quality.

KEPA/UNEP (2026). Section I - National Vision, Objectives and Pillars of Environmental Data Governance.

KEPA/UNEP (2026). Section II - Alignment with Key National and International Frameworks.

KEPA/UNEP (2026). Section III - Benchmarking Best Practices in Environmental Data Governance.

KEPA/UNEP (2026). Section IV Part I - Situational Analysis: Analysis of Current State of National Environmental Data.

KEPA/UNEP (2026). Section IV Part II - Situational Analysis Assessment of EPA NEDMP Updated Departments.

KEPA/UNEP (2026). Section IV Part III - Situational Analysis Assessment of All Stakeholders Environmental Data Management and Governance.

KEPA/UNEP (2026). Kuwait National Environmental Data Management Plan (NEDMP).

KEPA/UNEP (2026). NEDMP Data Classification Governance Framework.

SDMX Sponsoring Organizations (2025). SDMX Technical Specifications, version 3.1. Official SDMX standards page.

UN Statistics Division (2013). Framework for the Development of Environment Statistics.

United Nations et al. (2012). System of Environmental-Economic Accounting 2012 - Central Framework.

UNEP (2026). Global Environmental Data Strategy (GEDS). Official UNEP topic page.

Wilkinson, M. D. et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. Scientific Data.